

Fraud Transaction Detection (Linear Search)

```
import java.util.Scanner;
```

```
public class Main{
```

```
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter no. of transactions: ");  
        int n = sc.nextInt();  
        int[] transactions = new int[n];
```

```
        System.out.println("Enter transaction IDs: ");  
        for (int i = 0; i < n; i++){  
            transactions[i] = sc.nextInt();  
        }
```

```
        System.out.println("Enter target transaction ID: ");  
        int target = sc.nextInt();
```

```
        boolean found = false;  
        for (int i = 0; i < n; i++){  
            if (transactions[i] == target){  
                found = true;  
                break;  
            }  
        }
```

```
        if (found){  
            System.out.println("Fraud Transaction Found");  
        }  
        else{
```

```
System.out.println("Transaction Not  
Found");
```

```
}
```

```
}
```

```
}
```

Enter no. of transactions :

5

Enter transaction IDs :

1204

2345

7892

5611

1345

Enter target transaction ID :

5611

Fraud Transaction Found

2. Server Log Search (Binary Search)

```
import java.util.Scanner;
```

```
public class ServerLogSearch {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter no of logs : ");  
        int n = sc.nextInt();  
        int[] logs = new int[n];
```

```
        System.out.println("Enter log IDs : ");  
        for(int i=0; i<n; i++){  
            logs[i] = sc.nextInt();  
        }
```

```
        System.out.println("Enter target log ID: ");  
        int target = new sc.nextInt();
```

```
        int left=0, right=n-1;  
        boolean found=false;
```

```
        while (left <= right) {  
            int mid = (left+right)/2;
```

```
            if (logs[mid] == target) {  
                found = true;  
                break;
```

```
            }  
            else if (logs[mid] < target) {  
                left = mid+1;
```

```
}  
else {  
    right = mid - 1;  
}  
}  
  
if (found) {  
    System.out.println("Log Found");  
}  
else  
    System.out.println("Log Not Found");  
}  
}
```


Enter no. of logs :

5

Enter log IDs :

1002

1014

1078

1099

1103

Enter target log ID :

1002

Log Found

3. Sorting Salaries (Bubble Sort)

```
import java.util.Scanner;
```

```
public class BubbleSortSalaries{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter no. of employees : ");  
        int n = sc.nextInt();  
        int[] salaries = new int[n];
```

```
        System.out.println("Enter salaries : ");  
        for(int i=0; i<n; i++){  
            salaries[i] = sc.nextInt();  
        }
```

```
        for(int i=0; i<n-1; i++){  
            for(int j=0; j<n-i-1; j++){  
                if(salaries[j] > salaries[j+1]){  
                    int temp = salaries[j];  
                    salaries[j] = salaries[j+1];  
                    salaries[j+1] = temp;  
                }  
            }  
        }
```

```
        System.out.println("Sorted Salaries : ");  
        for(int i=0; i<n; i++){  
            System.out.print(salaries[i] + " ");  
        }  
    }  
}
```

Enter no. of employees :

5

Enter salaries :

45000

20000

55000

72000

35000

Sorted Salaries :

20000	35000	45000	55000	72000
-------	-------	-------	-------	-------

4. Sorting Response Time (Insertion Sort)

```
import java.util.Scanner;

public class ResponseTimeSort {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter total no. of Response Times : ");
        int n = sc.nextInt();
        int[] times = new int[n];

        System.out.println("Enter time : ");
        for (int i = 0; i < n; i++) {
            times[i] = sc.nextInt();
        }

        for (int i = 1; i < n; i++) {
            int key = times[i];
            int j = i - 1;

            while (j >= 0 && times[j] > key) {
                times[j + 1] = times[j];
                j--;
            }

            times[j + 1] = key;
        }

        System.out.println("Sorted Response Time : ");
    }
}
```

Date _____

Page _____

```
for(int i=0; i<n; i++){  
    System.out.print(times[i]+" ");  
}
```

```
}
```

Enter total no. of Response Times :

5

Enter time :

250

130

200

415

100

Sorted Response Time :

100	130	200	250	415
-----	-----	-----	-----	-----

5. Sorting Priorities (Selection Sort)

```
import java.util.Scanner;
```

```
public class PrioritySort{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter no. of Bugs : ");
```

```
        int n = sc.nextInt();
```

```
        int[] priority = new int[n];
```

```
        System.out.println("Enter Priority : ");
```

```
        for(int i=0; i<n; i++){
```

```
            priority[i] = sc.nextInt();
```

```
        }
```

```
        for(int i=0; i<n-1; i++){
```

```
            int minIndex = i;
```

```
            for(int j=i+1; j<n; j++){
```

```
                if (priority[j] < priority[minIndex]){
```

```
                    minIndex = j;
```

```
                }
```

```
            }
```

```
            int temp = priority[i];
```

```
            priority[i] = priority[minIndex];
```

```
            priority[minIndex] = temp;
```

```
        }
```

```
System.out.println("Sorted Priority: ");  
for(int i=0; i<n; i++){  
    System.out.print(priority[i] + " ");  
}
```

```
}
```


Enter no. of Bugs :

5

Enter Priority :

2

1

6

9

4

Sorted Priority :

1

2

4

6

9